

CSA0720-MCQ

Total points 9/20

C03-AT3

M. Guocharan.

192524434.

✓ 18 Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases *1/1

- ☐ a) Cache hit ratio
- ☒ b) Recursive resolver query load
- ☐ c) DNSSEC signature validity
- ☐ d) Zone transfer frequency

✓ 11. Head-of-line blocking in HTTP/1.1 occurs when; * 1/1

- ☐ a) The DNS server is slow
- ☒ b) A response blocks subsequent responses on the same connection
- ☐ c) TCP CWND drops to zero
- ☐ d) The server rejects pipelined requests

✗ 2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes: *0/1

- ☐ a) 9 MSS
- ☒ b) 16 MSS
- ☐ c) 12 MSS
- ☐ d) 8 MSS

Correct answer

- ☒ a) 9 MSS

✗ 8. What is the maximum UDP payload size without IP-level fragmentation *0/1 on a standard Ethernet link (MTU = 1500 bytes)?

- ☒ a) 1460 bytes
- ☐ b) 1472 bytes
- ☐ c) 1460 bytes
- ☐ d) 1500 bytes

Correct answer

- ☒ b) 1472 bytes

✗ 9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is: *0/1

- ☒ a) Eliminating IP header overhead
- ☐ b) 0-RTT or 1-RTT connection setup
- ☐ c) Removing encryption
- ☐ d) Native multicast support

Correct answer

- ☒ b) 0-RTT or 1-RTT connection setup

✓ 12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at *1/1 the:

- ☐ a) Application layer

9/20
45%

19. Which of the following is not a symptom of a network congestion problem?
a) Increased latency
b) Increased packet loss
c) Increased throughput
d) Increased jitter

☒ a) Increased latency

☐ b) Increased packet loss

☐ c) Increased throughput

☐ d) Increased jitter

Correct answer

☒ b) Increased packet loss

20. Which of the following is not a symptom of a network congestion problem?
a) Increased latency
b) Increased packet loss
c) Increased throughput
d) Increased jitter

☒ a) Increased latency

☐ b) Increased packet loss

☐ c) Increased throughput

☐ d) Increased jitter

Correct answer

☒ b) Increased packet loss

21. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?
a) 301
b) 302
c) 304
d) 307

☐ a) 301

☒ b) 302

☐ c) 304

☐ d) 307

Correct answer

☒ a) 301

Name of the student Reg No *

GURUCHARAN IA-192524484

22. Which of the following is not a symptom of a network congestion problem?
a) Increased latency
b) Increased packet loss
c) Increased throughput
d) Increased jitter

☐ a) Increased latency

☒ b) Increased packet loss

☐ c) Increased throughput

☐ d) Increased jitter

23. Which field in the TCP header is used to implement flow control at the receiver side?
a) Sequence number
b) Acknowledgement number
c) Window size
d) Urgent pointer

☐ a) Sequence number

☒ b) Acknowledgement number

☐ c) Window size

☐ d) Urgent pointer

Correct answer

☒ c) Window size

✓ 12 HTTP/2 multiplexing reduces latency by sending all the HTTP layer traffic at 1/1
the

- ☐ a) Application layer
- ☐ b) DNS layer
- ☒ c) TCP layer
- ☐ d) IP layer

✓ 14 Under 80% packet loss, which protocol combination gives the best 4/1
performance for a real time multiplayer game?

- ☐ a) TCP for game state, HTTP for assets
- ☒ b) UDP with application layer sequencing for game state, HTTP/S for assets
- ☐ c) HTTP/S for all traffic
- ☐ d) TCP with Nagle enabled for all traffic

✗ 15 Which DNS record type maps a domain name to an IPv6 address? 0/1

- ☐ a) A
- ☒ b) CNAME
- ☐ c) AAAA
- ☐ d) MX

Correct answer

- ☒ c) AAAA

✓ 17 Which attack exploits the UDP-based DNS protocol by sending forged 4/1
responses with a matching transaction ID?

- ☐ a) ARP poisoning
- ☒ b) DNS cache poisoning
- ☐ c) SYN flooding
- ☐ d) BGP hijacking

✓ 16 A DNS resolver switches from UDP to TCP when: 1/1

- ☐ a) The query type is MX
- ☒ b) The response is truncated (TC bit set)
- ☐ c) TTL exceeds 3600 s
- ☐ d) The authoritative server is unreachable

✗ 7 A UDP-based streaming application experiences 15% packet loss. The 4/1
best application layer mitigation is:

- ☐ a) Increase UDP buffer size
- ☐ b) Switch to TCP
- ☐ c) Implement Forward Error Correction (FEC)
- ☒ d) Reduce frame rate to 1 fps

Correct answer

- ☒ c) Implement Forward Error Correction (FEC)

✓ 10 HTTP/1.1 persistent connections reduce latency by: 1/1

✓ 10. In TCP, persistent connections reduce latency by:

1/1

- ☐ a) Forwarding RST in the end
- ☒ b) Reusing the same TCP connection for multiple requests
- ☐ c) Multiplexing HTTP requests
- ☐ d) Sending requests without waiting for reply

✗ 11. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?

0/1

- ☒ a) Slow Start
- ☐ b) Fast Retransmit
- ☐ c) Fast Recovery
- ☐ d) Tahoe Reset

Correct answer:

- ☒ c) Fast Recovery

✗ 12. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

0/1

- ☒ a) MTU size
- ☐ b) CWND growth rate relative to RTT
- ☐ c) DNS resolution time
- ☐ d) UDP competition

Correct answer:

- ☒ b) b) CWND growth rate relative to RTT

✓ 13. UDP is preferred over TCP for DNS queries primarily because:

1/1

- ☐ a) UDP supports fragmentation
- ☒ b) DNS responses are small and connection overhead is unnecessary
- ☐ c) TCP cannot use port 53
- ☐ d) UDP has better error correction

✗ 14. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

0/1

- ☒ a) HTTP, TCP, DNS
- ☐ b) DNS, TCP handshake, HTTP GET
- ☐ c) TCP, DNS, HTTP
- ☐ d) DNS, HTTP, TCP

Correct answer:

- ☒ b) b) DNS, TCP handshake, HTTP GET

✗ 15. In HTTP caching, the Cache-Control: no-cache directive means:

0/1

- ☒ a) Never cache this resource
- ☐ b) Validate with the server before using a cached copy
- ☐ c) Cache for 0 seconds
- ☐ d) Disable all caching headers

Correct answer:

- ☒ b) b) Validate with the server before using a cached copy